

Name-Mujahid

USN-2VX24UE031

Expt.No-3

Title-merge Sort

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In [5]: 1 import random
2 import time
3 import matplotlib.pyplot as plt
4
5 def merge_sort(arr):
6     if len(arr) <= 1:
7         return arr
8     mid = len(arr)//2
9     left = merge_sort(arr[:mid])
10    right = merge_sort(arr[mid:])
11    return merge(left, right)
12
13 def merge(left, right):
14     result = []
15     i = j = 0
16
17     while i < len(left) and j < len(right):
18         if left[i] < right[j]:
19             result.append(left[i])
20             i += 1
21         else:
22             result.append(right[j])
23             j += 1
24     return result + left[i:] + right[j:]
25
26 # Input from user
27 arr = list(map(int, input("Enter elements separated by space: ").split()))
28 print("Input array:", arr)
29 start_time = time.time()
30 sorted_arr = merge_sort(arr)
31 end_time = time.time()
32 print("Sorted array:", sorted_arr)
33 print("Time taken:", end_time - start_time, "seconds")
34 print("=====")
35 # For n > 5000
36
37 n_values = [5000, 6000, 7000, 8000, 9000]
38 time_values = []
39 for n in n_values:
40     arr = [random.randint(1, 9) for _ in range(n)]
41     start_time = time.time()
42     merge_sort(arr)
43     end_time = time.time()
44     time_taken=end_time-start_time
45     time_values.append(time_taken)
46     print("Time for", n, "element:",time_taken,"seconds")
47 print("=====")
48
49 # plotting the graph
50 plt.plot(n_values, time_values,color = 'black',marker = 'o')
51 plt.xlabel('Number of elements (n)')
52 plt.ylabel('Time taken (seconds)')
53 plt.title('merge sort time complexity analysis')
54 plt.grid(True)
55 plt.show()
56
```

Enter elements separated by space: 2 4 6 8 5

Input array: [2, 4, 6, 8, 5]

Sorted array: [2, 4, 5, 6, 8]

Time taken: 0.0 seconds

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Time for 5000 element: 0.007895469665527344 seconds

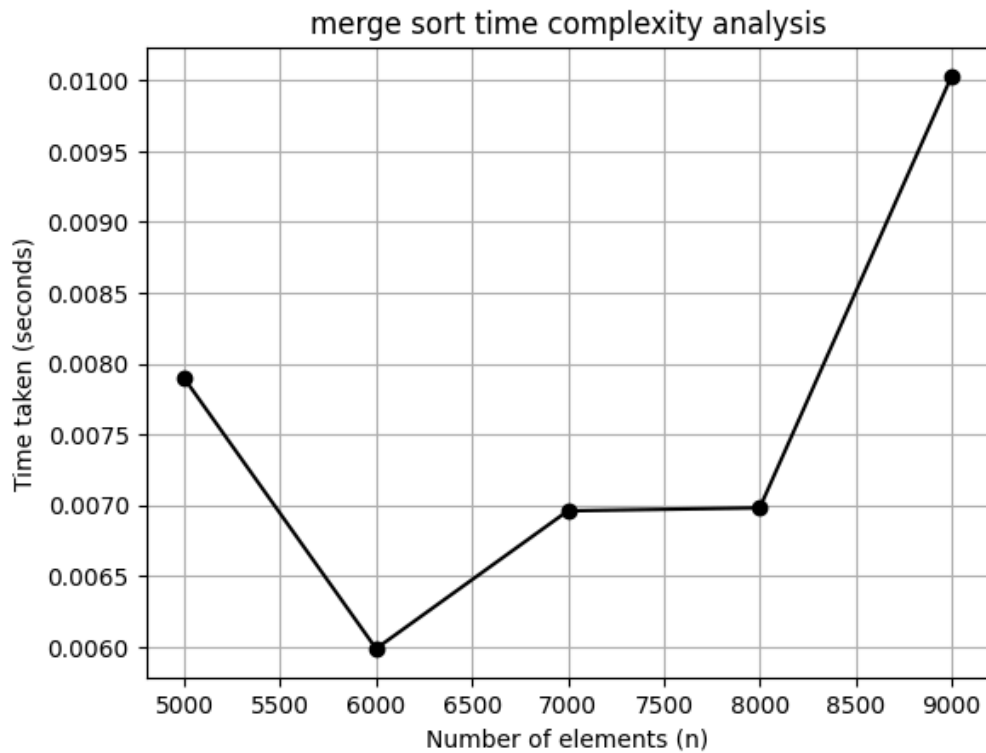
Time for 6000 element: 0.0059871673583984375 seconds

Time for 7000 element: 0.006960153579711914 seconds

Time for 8000 element: 0.0069811344146728516 seconds

Time for 9000 element: 0.010028362274169922 seconds

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In []:

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